

What is REAL right now

1. TCIR dataset — REAL ☒

You have real TCIR satellite data:

- 21,076 original records
- 10,538 cleaned samples
- 485 cyclones
- multiple basins

These aren't synthetic samples. [README](#)

2. TCIR intensity model — REAL + TRAINED ☒

This is one of the strongest completed parts.

Your temporal model actually predicts:

Satellite sequence



CNN



GRU



Wind

Pressure

Storm Size



using 4 chronological frames over ~9 hours. [README](#)

using 4 chronological frames over ~9 hours. [📄 README](#)

And you have held-out test results:

Target	MAE	RMSE	R ²
Wind	12.43 kt	16.91 kt	0.6769
Pressure	8.84 hPa	12.86 hPa	0.6944
Size	31.35 nmi	44.59 nmi	0.4606

Those results are specifically from the **TCIR test split**, not INSAT. [📄 README](#)

So when your UI shows:

Wind: 29.03 kt

Pressure: 1005.42 hPa

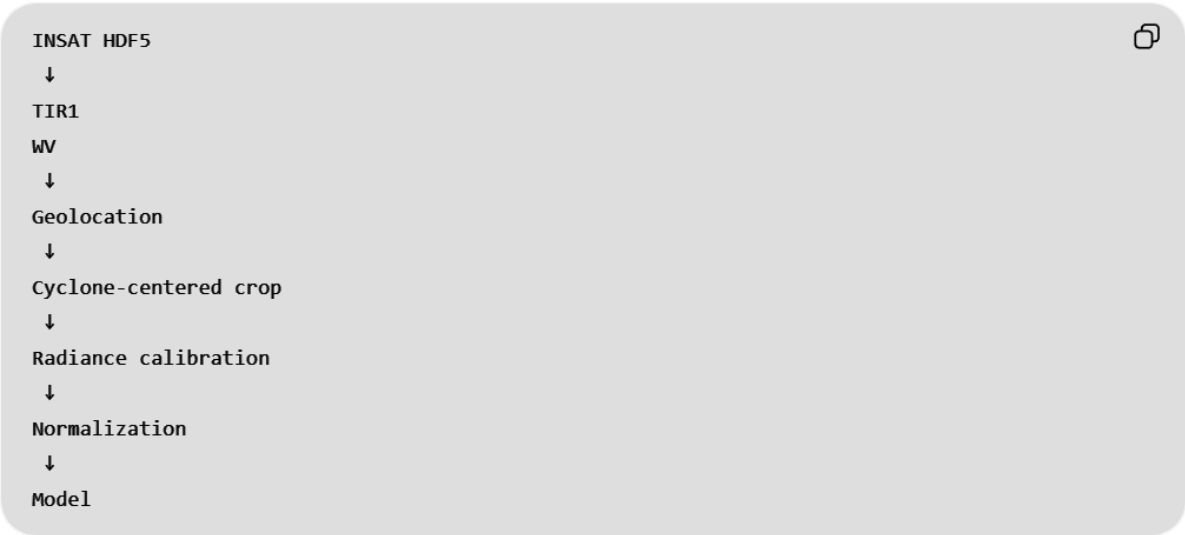
Size: 8.33 nmi

that's an actual inference from your TCIR model, assuming you're running the checkpoint currently connected to that endpoint.

3. INSAT — REAL DATA, but model is NOT finished ⚠

This part is important.

You really are using INSAT-3D data and MOSDAC processing. Your pipeline includes:



That's documented in the README. [README](#)

And you've already processed:

FANI	10 observations
TAUKTAE	12 observations
BULBUL	12 observations

TOTAL	34

But this is **NOT** yet:

"Give arbitrary INSAT image → reliably detect cyclone → predict intensity"

because you don't yet have enough independent NIO data and haven't completed held-out storm evaluation.

Your README explicitly says real-world cyclone-presence detection hasn't been validated.

 README

4. Synthetic model — NOT real cyclone intelligence ✕

This is the part you should **not** present as your actual cyclone prediction model.

You have:

```
classifier.pt
predictor.pt
```



but they're legacy synthetic models.

The README explicitly says synthetic performance should **not** be interpreted as real-world cyclone detection capability.

 README

So your UI should keep:

SYNTHETIC VALIDATION

and not:

REAL CYCLONE PREDICTION

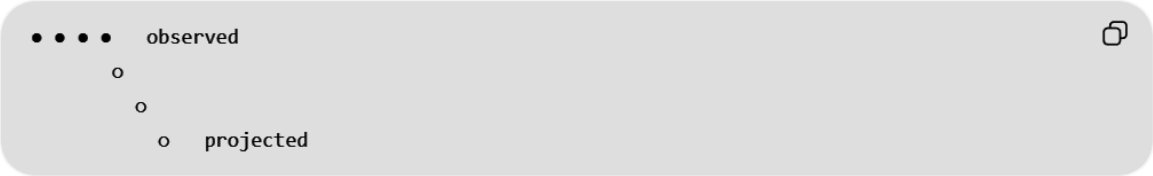
That's actually scientifically correct.



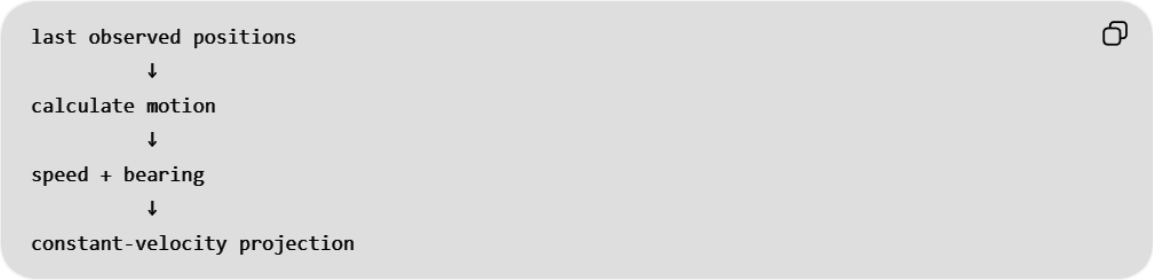
5. Track map — currently NOT AI track prediction ⚠

This is the biggest distinction.

Right now your nice map showing:



is useful visually, but the projected portion we added is essentially:



It is **not** your trained neural network predicting $\Delta\text{Lat}/\Delta\text{Lon}$.

Your README agrees with this: the actual track model is still a planned development stage. [README](#)

So currently:

Real

Observed track ☒

Calculated

Motion / heading / short-horizon extrapolation ☒ 

Not yet done

AI future track prediction 

PHASE A — Finish INSAT dataset

Current:

```
FANI
TAUKTAE
BULBUL
```



34 samples.

Need to properly consolidate:

```
FANI
TAUKTAE
BULBUL
AMPHAN
NISARGA
YAAS
GULAAB
ASANI
BIPARJOY
MICHAUNG
```



Your README already has these ten target storms.

 README

Then create something like:

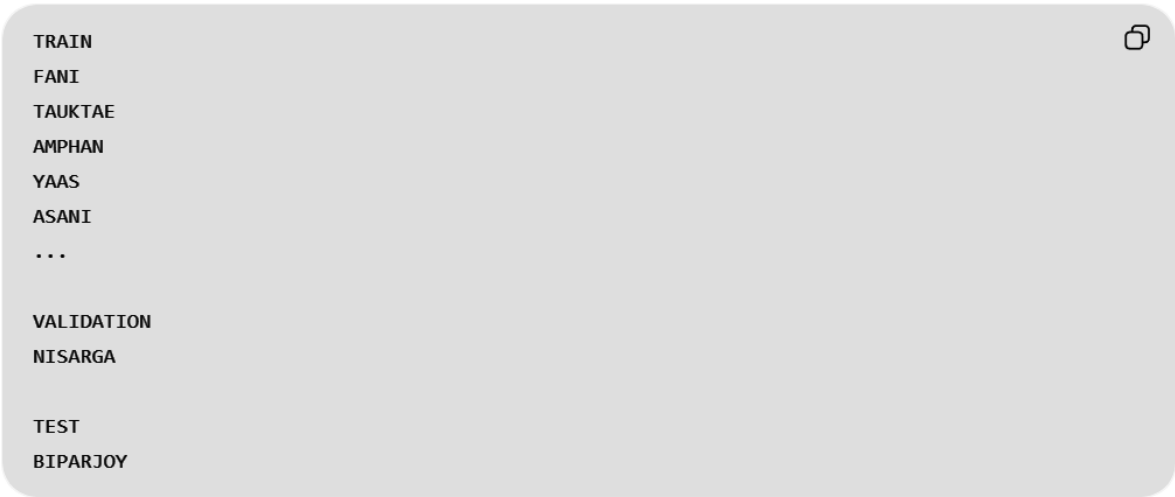
```
insat_combined/
|
├─ train/
├─ val/
└─ test/
```



But **split by storm**, not randomly by image.

But **split by storm**, not randomly by image.

For example:



```
TRAIN
FANI
TAUKTAE
AMPHAN
YAAS
ASANI
...

VALIDATION
NISARGA

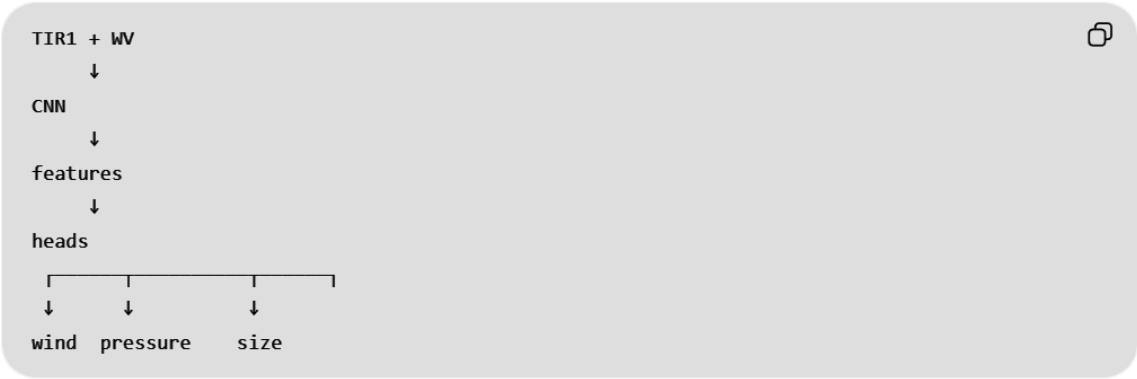
TEST
BIPARJOY
```

The README specifically calls for storm-level separation to avoid leakage between frames of the same cyclone. [📄 README](#)

PHASE B — Train proper INSAT model

Right now you have an INSAT checkpoint, but the README still marks proper NIO fine-tuning and held-out evaluation as unfinished.

The target should be:



Potentially classification as a separate head.

Then evaluate **only on storms the model never saw during training**.

You need a table like:

Storm	Wind MAE	Pressure MAE	Size MAE	
Held-out A	...	...	...	
Held-out B	...	...	...	
Overall	...	...	...	

That becomes your real NIO result.



PHASE C — Actual AI track prediction

This is the biggest remaining ML feature.

Currently:

```
Latt  
Lont  
Latt-1  
Lont-1
```



→ motion calculation.

Instead we need training sequences:

```
t-12h  
t-9h  
t-6h  
t-3h  
t
```



with:

```
lat  
lon  
wind  
pressure  
satellite features
```



Then target:

Δlat
 Δlon



For example:

INPUT



t-12h	10.2N	85.1E
t-9h	10.5N	85.5E
t-6h	10.8N	85.9E
t-3h	11.1N	86.3E
t	11.4N	86.7E

↓

CNN + GRU / LSTM

↓

Predicted $\Delta Lat = +0.4^\circ$
Predicted $\Delta Lon = +0.5^\circ$

Then:

future position =
current position + predicted displacement



That's the actual feature described in your README. [README](#)

PHASE D — Proper evaluation

This is where VayuNetra becomes much more defensible.

For track:

Latitude MAE

Longitude MAE

Position RMSE

Great-circle distance

Track error km

Track error km/h

Those are already specified in your README.

README

Then your final result can say something like:

TCIR

Wind MAE12.43 kt

Pressure MAE8.84 hPa

INSAT/NIO

Wind MAEX kt

Pressure MAEY hPa

TRACK

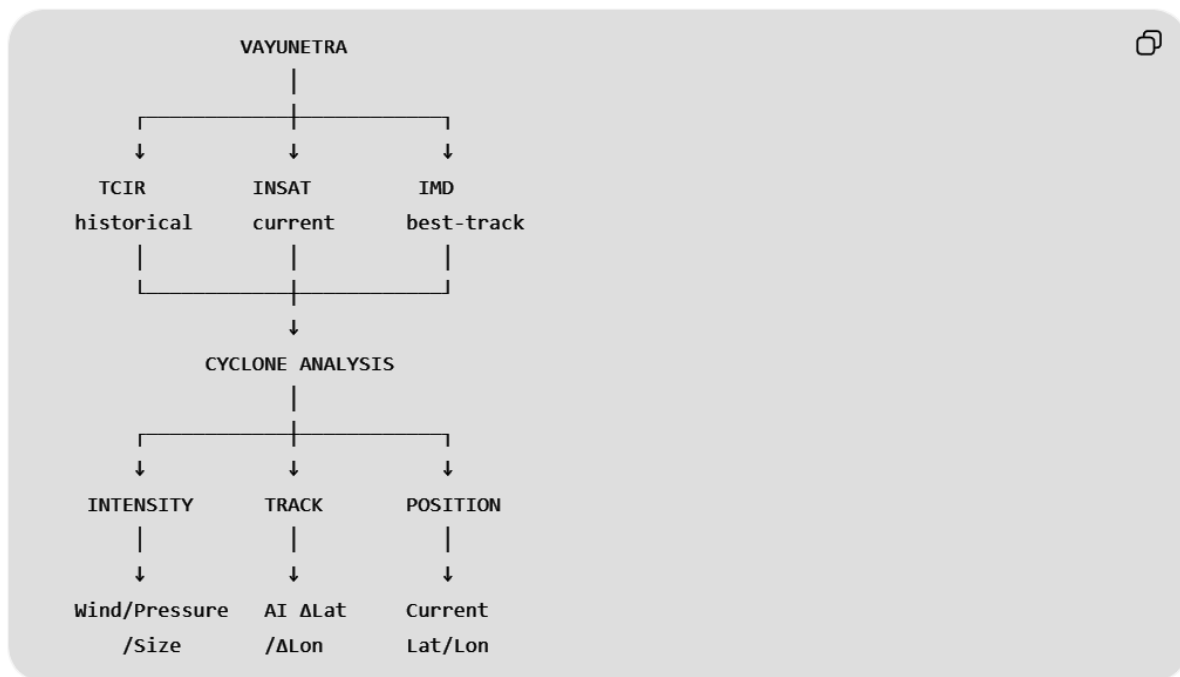
Mean errorX km

24h errorY km

Only after actually running those evaluations.

And then your frontend becomes genuinely meaningful

Your current UI can eventually represent:



Then the frontend isn't just decoration.

It's displaying outputs from actual pipelines.

REAL TCIR DATA	✓
TCIR INTENSITY MODEL	✓
TCIR TEMPORAL MODEL	✓
IMD BEST TRACK	✓
INSAT DATA	✓
INSAT PREPROCESSING	✓
34 REAL INSAT OBSERVATIONS	✓
FASTAPI BACKEND	✓
FRONTEND	✓
TRACK VISUALIZATION	✓



The unfinished scientific pieces are:

10-storm NIO dataset	●
INSAT fine-tuning	●
held-out NIO evaluation	●
AI track model	●
track evaluation	●
final end-to-end validation	●

